StudifyX

An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

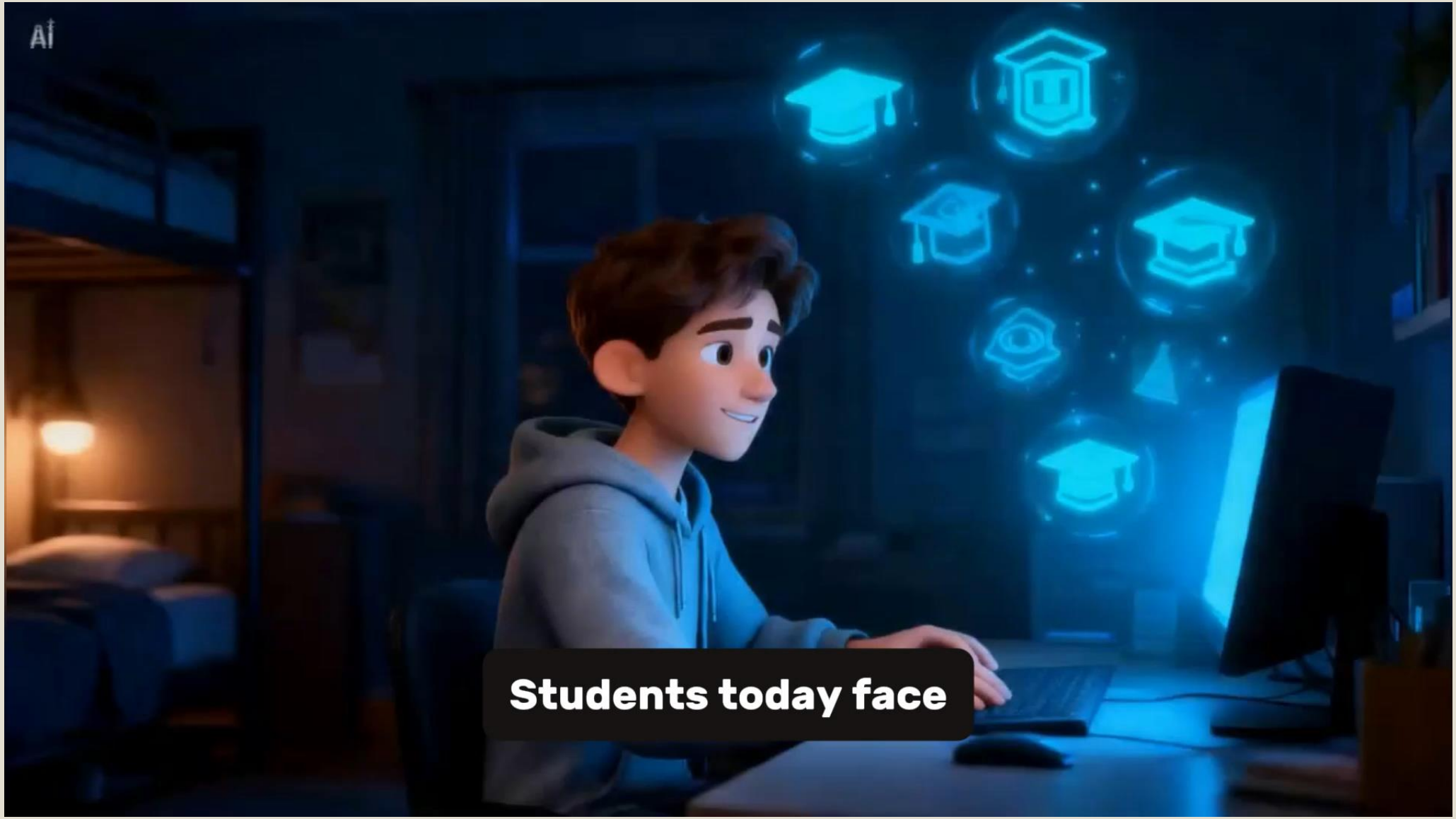
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R26-IT-073


AI
Students today face

Problem & Student Pain Points

1. Blind Degree Decisions

Students pick university degrees blindly without understanding long-term career demands or Z-score requirements.

2. Industry Skill Mismatch

most of the candidate profiles lack the exact technical competencies demanded by current job roles.

3. Unverified Mentorship

Existing guidance channels lack credential checks, leading to unreliable and unverified career advice.

4. Academic Burnout & Stress

Students struggle with heavy exam workloads and fatigue without intelligent, personalized study balancing.

Research Gap & Project Objectives

Existing Tools Limitation

- **Siloed Systems:** Degree tools, job boards, and calendars operate in complete isolation.
- **Unquantified Skill Deficiencies:** Standard job matching platforms fail to analyze or measure the exact skill gaps between a user's current competencies and industry standards.
- **Zero Verification:** No automated CV/certificate auditing for career mentors.

StudiFyX Objectives

- **Backward Chaining:** Map degrees directly from dream job goals.
- **Skill-Gap Analyzer (SGI):** Quantifies the semantic distance between the candidate profile and target industry role to generate a personalized learning roadmap.
- **Automated Verification:** OCR CV analysis and structured scoring.
- **Bio-Adaptive Schedule:** Smart study plans balancing stress and fatigue.

The Four Core Pillars of StudiFyX

1. Dream Degree Advisor

Backward-chains from student dream goals, A/L results, and Z-scores to recommend optimal degrees and universities.

2. Career path & Skill-Gap Analyzer

Calculates real-time Skill-Gap Index (SGI) using ML ensembles and Neo4j graph ontology for prescriptive roadmaps.

3. Mentor Hub & Network

Verifies mentors via OCR CV parsing, automated interview evaluation, and assigns Bronze/Silver/Gold trust badges.

4. Cognitive Wellbeing AI

Monitors student workload and fatigue using Random Forest AI to dynamically schedule adaptive study sessions.

RESEARCH CONTRIBUTION

Design Excellence & Unique Research Value

First Integrated Ecosystem

Only platform combining degree mapping, skill-gap analysis, mentor validation, and study wellbeing in a single unified system.

Backward Career Chaining

Reverses traditional academic matching by starting from dream career goals and mapping required degrees and universities.

Mathematical SGI Metric

Quantifies exact semantic competency distances using Cosine Similarity and Neo4j dynamic knowledge graph ontologies.

Automated Trust Auditing

Eliminates unverified advice through OCR CV extraction and automated interview scoring to assign certified mentor badges.

Bio-Adaptive Wellbeing AI

Prevents student burnout by dynamically adjusting study sessions using Random Forest AI (94.5% accuracy) and telemetry.

Sri Lankan Localization

Includes trilingual support (English, Sinhala, Tamil), Singlish preprocessing, and national Z-score cutoffs.

Component Integration & Readiness

Sub-System Component	Lead Member	Key Technology	Status
Dream Degree Advisor	Rusiru L.W.S (IT22220288)	Degrees and Universities Recommendation, Backward Chaining & Multi-Parameter Rankings	Completed / Integrated
Career & Skill-Gap Analyzer	Wickremathilaka J.A (IT22152114)	Stacking Ensemble & Cosine SGI Formula	Completed / Integrated
Mentorship & Alumni Network	Mahikalpa K.B.V.I (IT22150370)	OCR CV Parsing & Automated Interview AI	Completed / Integrated
Smart Study & Stress AI	Withanage D.K (IT22094636)	Random Forest Workload & Stress Estimator	Completed / Integrated

Market Opportunity & Sustainability

Target Market Size

- **250,000+ A/L Students:** Seeking annual university & degree guidance.
- **Students & Any Career Seekers:** Undergraduates, job hunters.
- **Trilingual Reach:** Fully localized for Sinhala, Tamil, and English users.

Sustainable Revenue Streams

- **B2B University Licensing:** Curriculum gap analytics dashboard for higher education.
- **Freemium & Micro-Reports:** Free core features with 450–750 LKR deep-dive reports.
- **Verified Mentor Ecosystem:** Premium counseling sessions & institutional training links.

Commercialization & Service Model

- **The proposed system can be commercialized through a three-tier service model.**

Basic (Freemium)

Free

- Backward Dream Degree discovery
- High-level Career Path prediction
- General university entry cutoffs
- Community mentor directory access

For general school leavers

Pro Student

450 - 750 LKR

- All Basic features included
- Full SGI Skill-Gap breakdown
- Neo4j prescriptive learning roadmap
- Bio-Adaptive Pomodoro & study planner
- Downloadable readiness PDF report

Institutional / B2B

Annual License

- University career guidance portal
- Cohort-wide skill deficit analytics
- Verified mentor verification portal

For universities & institutes

Targeted IT Course & Job Ads

Verified Mentor Session Commission

Industry Talent Insights Dashboard

Live Demo Evaluation Roadmap

1. Degree Adviser

Input A/L results & dream goal to observe backward-chained degree recommendations.

2. Career path &Skill Gap Analyzer

View SGI score and interactive skill roadmap powered by Neo4j graph.

3. Professional Mentorship and Alumni Network

Demonstrate live CV OCR parsing and badge-tier verification matching.

4. Real-time Stress Management And Mental Wellbeing

Simulate workload inputs to trigger real-time adaptive study schedule balancing.

Thank You.